

# **Literature Review**

## **Machine Learning-Based Predictive Irrigation System for Smart Water Gate Control in Paddy Cultivation**

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RGU Number: 2313322

IIT Number: 20221647

October, 23, 2025

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## 1 Introduction

Efficient water management in agriculture remains a critical challenge in the context of climate variability, population growth, and sustainable resource utilisation. Paddy cultivation, in particular, demands significant water input due to its continuous flooding requirement, making it one of the most water-intensive crops globally. Traditional irrigation scheduling methods continue to rely heavily on farmer intuition and fixed-interval watering, which often leads to either over-irrigation or water stress. These inefficiencies highlight the pressing need for intelligent, data-driven irrigation systems that integrate prediction, control, and automation.

This chapter provides a critical review of existing research and developments underpinning the proposed *Smart Water Gate System for Paddy Cultivation*. The review situates the project within the broader fields of precision agriculture, artificial intelligence (AI), and Internet of Things (IoT)-based automation. It systematically explores the theoretical and conceptual background, the methods and technologies adopted in prior work, and related empirical studies in irrigation automation, soil-moisture estimation, and crop-water requirement modelling. Finally, it identifies significant research gaps that motivate the proposed study.

The chapter is structured as follows. Section 2 outlines the theoretical and conceptual foundations relevant to intelligent irrigation and predictive water management. Section 3 reviews key methods, algorithms, and tools used in similar domains. Section 4 examines prior studies and their findings, highlighting both their contributions and limitations. Section 5 synthesises the identified research gaps and justifies the novelty of this study. Section 6 concludes the review with a summary of key insights that inform the subsequent research methodology.

## 2 Theoretical and Conceptual Background

The foundation of this research is anchored on four key theoretical pillars: **Precision Agriculture**, **Cyber-Physical Systems (CPS)**, **Machine Learning (ML)**, and **Control Theory**. These domains collectively underpin the conceptual architecture of the proposed *Machine Learning-Based Predictive Irrigation System for Smart Water Gate Control in Paddy Cultivation*. Each contributes to transforming irrigation from a reactive, human-dependent process to a proactive, data-driven, and autonomous framework tailored for paddy environments.

### 2.1 Precision Agriculture and Smart Irrigation

Precision agriculture embodies the integration of spatial and temporal variability in crop and soil conditions to optimise input use and enhance productivity. In irrigation management, this translates to delivering the right amount of water at the right place and time by leveraging continuous environmental monitoring. The principle aims to replace fixed-interval or intuition-based irrigation scheduling with data-driven decision-making.

In paddy cultivation, where water plays a critical physiological and ecological role, precision irrigation ensures optimal soil saturation while conserving limited water resources. Studies such as [Arif et al. \(2013\)](#) demonstrated that data-oriented models using meteorological parameters—temperature, precipitation, and evapotranspiration—can effectively estimate soil moisture without requiring complex hydrological equations. Similarly, [Togneri et al. \(2022\)](#) established a comprehensive machine-learning framework that forecasts soil-moisture variation across different crop and climatic conditions, emphasising the predictive potential of high-frequency data.

Satellite and remote-sensing technologies have also contributed to precision irrigation by providing macro-level soil-moisture and salinity mapping. For instance, [Ghazali et al. \(2019\)](#) and [Sholihah et al. \(2022\)](#) employed Landsat and optical imagery to monitor field conditions over large areas. While these approaches offer wide coverage, they lack the real-time adaptability needed for responsive irrigation control. IoT-based systems, such as those introduced by [Nguyen et al. \(2022\)](#), bridge this gap through multi-sensor fusion, combining ground-level data with satellite information for higher accuracy and resolution.

By integrating these sensing and forecasting capabilities, precision agriculture provides the empirical and methodological foundation for developing predictive and autonomous irrigation systems—such as the one proposed in this study—that dynamically balance crop water needs and environmental constraints.

## 2.2 Cyber-Physical Systems (CPS)

A Cyber-Physical System (CPS) represents a holistic framework that fuses computational intelligence with physical processes through sensors, embedded processors, actuators, and communication networks. In agricultural applications, CPS enables seamless interaction between environmental data acquisition, algorithmic analysis, and automated actuation in real time.

Within irrigation, CPS acts as the “nervous system” of smart water-management infrastructure. Its architecture typically comprises a perception layer (sensor network for soil, rainfall, and water-level data), a communication layer (IoT-based transmission), a processing layer (edge or cloud-based analytics), an actuation layer (motorised gates or pumps), and an application layer (user interface and monitoring). [Rathnayaka and Park \(2024\)](#) highlighted this CPS perspective in their review of automatic gate-control systems for sustainable agriculture, describing how integrated sensors, microcontrollers, and solar-powered actuators can reduce manual labour and improve water allocation efficiency.

Similarly, [Rama et al. \(2021\)](#) demonstrated that embedded controllers can autonomously regulate irrigation gates using fuzzy logic rules within a CPS-like feedback loop. Meanwhile, [Niharika et al. \(2024\)](#) extended this concept through GSM-based monitoring, enabling farmers to supervise irrigation remotely. These developments collectively reinforce the CPS notion of connected intelligence—where computational decision-making and physical processes coexist

as an integrated system.

The proposed smart water gate system adopts this CPS paradigm to link sensory data, predictive modelling, and mechanical control within a unified feedback architecture. Such integration enables the system to continuously perceive environmental conditions, compute optimal irrigation strategies, and execute gate adjustments with minimal human intervention.

### 2.3 Machine Learning and Predictive Modelling

Machine learning contributes the analytical core to intelligent irrigation, enabling predictive rather than reactive management. By learning from historical and real-time sensor data, ML algorithms can capture nonlinear relationships between variables such as rainfall, temperature, evapotranspiration, and soil moisture—allowing the system to forecast future water requirements with high precision.

[Arif et al. \(2013\)](#) pioneered this idea by employing Artificial Neural Networks (ANNs) to estimate soil-moisture content using basic climatic inputs, achieving robust predictive performance. Later works such as [Chatterjee et al. \(2020\)](#) and [Prasad et al. \(2018\)](#) improved model stability by hybridising ANNs with metaheuristic optimisers like the Firefly and Extreme Learning Machine algorithms. These studies underscored the potential of ML for modelling soil-water dynamics without dependency on physical hydrological simulations.

More recently, ensemble and automated learning frameworks have achieved near-real-time estimation accuracy. [Babaeian et al. \(2021\)](#) applied AutoML to fuse ground and remote-sensing data for root-zone soil-moisture estimation, while [Karthikeyan and Mishra \(2021\)](#) extended ML models across multiple soil layers to enhance spatial resolution. Additionally, deep learning approaches such as the BCO-ResNet model proposed by [Anbarasu and Parthipan \(2025\)](#) have advanced irrigation prediction by combining optimisation algorithms with residual networks, reducing water use by over 38%.

At the crop-requirement level, ML models have also proven effective in estimating and forecasting CWRs. [Nagappan et al. \(2025\)](#) demonstrated that LSTM, RF, and ANN architectures can achieve up to 0.92 accuracy in predicting irrigation needs. These studies collectively justify the adoption of ML as the central predictive mechanism in this research, providing the foresight necessary for intelligent water-gate control under dynamic environmental conditions.

### 2.4 Control Theory and Embedded Automation

While ML provides predictive intelligence, control theory governs the physical implementation of irrigation decisions. Classical control methods such as Proportional-Integral-Derivative (PID) and fuzzy logic controllers are widely used to regulate water flow and maintain desired water levels in irrigation channels. For example, [Sanjula et al. \(2020\)](#) implemented a PID-based

IoT system that automatically adjusts gate openings to maintain a fixed water depth, demonstrating effective yet static control. Similarly, [Rama et al. \(2021\)](#) developed a fuzzy logic controller capable of determining gate positions based on soil-moisture and water-level readings, offering robustness under variable field conditions.

However, both approaches remain reactive—they respond to real-time measurements but do not anticipate changes or learn from historical patterns. Studies such as [Desnanjaya and Nugraha \(2021\)](#) and [Tolentino et al. \(2021\)](#) further reveal the prevalence of web-based and GSM-controlled gate systems that still require manual oversight. Even modern Alternate Wetting and Drying (AWD) techniques evaluated by [Bwire et al. \(2024\)](#) rely on manual intervention, lacking sensor feedback or autonomous scheduling.

To overcome these limitations, the integration of ML prediction with control logic represents a next-generation paradigm. For instance, the predictive component can estimate upcoming soil-moisture decline, and the controller can pre-emptively regulate gate openings, achieving true closed-loop automation. Embedded devices such as Raspberry Pi or ESP32 facilitate this hybrid approach by running lightweight inference models locally, reducing latency and cloud dependence.

This synthesis aligns closely with the CPS model discussed earlier: perception through sensors, cognition via ML, and actuation governed by control theory. The resulting system not only reacts to present conditions but also anticipates future states, ensuring both precision and efficiency in water management.

### 3 Review of Key Methods and Technologies

A variety of computational, sensing, and automation technologies underpin intelligent irrigation research. This section critically evaluates major modelling techniques, data acquisition methods, and control mechanisms, identifying their contributions and limitations in paddy irrigation contexts.

#### 3.1 Soil-Moisture Estimation and Prediction Models

Accurate soil-moisture estimation forms the backbone of smart irrigation systems. [Arif et al. \(2013\)](#) demonstrated that ANNs could reliably estimate paddy-field soil moisture using meteorological inputs—temperature, precipitation, and evapotranspiration—achieving  $R^2 \approx 0.8$ . This data-driven approach eliminated complex physical simulations, paving the way for optimisation-enhanced models.

Building on this foundation, [Chatterjee et al. \(2020\)](#) introduced an ANN trained using Modified Flower Pollination Algorithm (MFPA) for stable moisture prediction under fluctuating conditions, while [Prasad et al. \(2018\)](#) proposed hybrid Extreme Learning Machine (ELM) frame-

works with Empirical Mode Decomposition to capture temporal soil-moisture behaviour, substantially reducing RMSE. These hybridisation strategies demonstrated that preprocessing and signal decomposition enhance learning outcomes.

Multi-source approaches further extended predictive capabilities. [Karthikeyan and Mishra \(2021\)](#) developed an XGBoost pipeline predicting soil moisture across five depths with unbiased RMSE below 0.04 m<sup>3</sup>/m<sup>3</sup>, while [Babaeian et al. \(2021\)](#) combined ground-based and remotely sensed data with AutoML, producing high-resolution root-zone moisture maps. Despite these advances, models remained primarily analytical without real-time control integration.

Deep learning has enhanced sensor calibration and spatial interpretation. [Kurtulmuş et al. \(2022\)](#) employed five CNN architectures—AlexNet, GoogleNet, ResNet, VGG16, and SqueezeNet—achieving F1-score of 0.9973 for irrigation requirement detection. However, testing occurred under controlled environments without field-level actuation integration. The proposed system addresses this gap through embedded predictive intelligence and real-time control.

### 3.2 IoT-Based Sensing and Data Fusion

Low-cost IoT sensors have revolutionised environmental monitoring, enabling real-time data collection. [Togneri et al. \(2022\)](#) proposed a machine learning-integrated IoT architecture using LightGBM and ensemble models for soil-moisture forecasting, demonstrating superior performance over traditional baselines. [Nguyen et al. \(2022\)](#) introduced multi-sensor fusion using Sentinel-1, Sentinel-2, and ALOS DSM data with GA-optimised XGBoost, achieving  $R^2 = 0.891$  at 10 m resolution. This showed that combining IoT and remote-sensing data enables both local precision and broader generalisation.

[Sholihah et al. \(2022\)](#) confirmed that optical remote sensing effectively estimates soil-moisture conditions in paddy fields, illustrating that ground-satellite sensor fusion provides holistic field dynamics understanding. However, these implementations lack feedback mechanisms for dynamic real-time irrigation adjustment. The proposed research bridges this gap by integrating IoT sensor data with predictive modelling and mechanical actuation in a closed-loop CPS configuration.

### 3.3 Automation and Control Systems

Automation transforms analytical insights into physical actions. [Sanjula et al. \(2020\)](#) introduced an IoT-based water-gate system using ESP-NOW and PID controllers to maintain predefined water levels, improving response time but requiring user configuration without adaptive intelligence. [Rama et al. \(2021\)](#) applied fuzzy logic control on ESP32 microcontrollers for automated sluice gates based on ultrasonic feedback, maintaining target depths without historical learning or forecast awareness.

Desnanjaya and Nugraha (2021) designed web-based control interfaces using Arduino and GSM modules for real-time monitoring but retained human supervision, while Niharika et al. (2024) introduced "Gate Guard" with cloud-GSM connectivity requiring manual SMS-triggered operations. These systems show incremental progress toward automation but lack adaptive, learning-enabled control.

Rathnayaka and Park (2024) reviewed automatic gate-controlling systems in Sri Lanka, emphasising self-regulating designs for varying hydrological conditions. Tolentino et al. (2021) implemented solar-powered IoT solutions applying AWD technique for improved water efficiency but maintained threshold-based operation. These studies reinforce the necessity for coupling automation hardware with intelligent, data-driven decision-making—the core objective of the proposed Smart Water Gate system.

### 3.4 Crop Water Requirement and Hydrological Modelling

Accurate CWR estimation underpins predictive irrigation scheduling. FAO CROPWAT-8.0 Singh and Verma (2020) uses Penman–Monteith formula and fixed crop coefficients for monthly irrigation schedules but lacks real-time adaptability. Awad et al. (2021) compared CROPWAT with DRAINMOD hydrological model, revealing up to 59% deviations in water requirement estimates, highlighting uncertainty in static modelling.

Machine learning increasingly overcomes these limitations. Nagappan et al. (2025) used ensemble ML models (LSTM, RF, SVM, ANN) achieving  $R^2 = 0.92$  for CWR forecasting. Anbarasu and Parthipan (2025) introduced hybrid BCO-ResNet deep learning framework reducing water use by 38% with 91.3% accuracy, though cloud-dependence constrains rural applicability. Additionally, Bwire et al. (2024) demonstrated AWD method achieving 29.2% water reduction when it compares to Continuous Flooding(CF)—but executed manually without automation or predictive feedback. Integrating such hydrological principles with automated CPS designs could yield environmental and operational benefits.

### 3.5 Advanced AI and Reinforcement Learning Approaches

Emerging research explores adaptive, learning-based control. Saikai et al. (2023) implemented DRL framework with APSIM crop simulator for optimised irrigation scheduling, autonomously learning water allocation strategies balancing conservation and yield. However, simulation-based approaches limit real-world sensor network transferability.

The convergence of ML prediction, control theory, and IoT actuation suggests hybrid, forecast-driven CPS architectures offer promising pathways for sustainable irrigation. The proposed system operationalises this concept by embedding predictive models and real-time decision logic within field-deployed hardware, creating scalable, autonomous water management frameworks tailored for paddy cultivation.



## 4 Review of Related Studies and Applications

This section reviews empirical and applied studies relevant to soil-moisture prediction, IoT-based irrigation automation, and crop water requirement forecasting, culminating in the identification of technological gaps addressed by the proposed Smart Water Gate System. The discussion synthesises advances across four interconnected domains: soil and environmental prediction studies, IoT and automation applications, crop water requirement forecasting, and hybrid AI-driven irrigation systems.

### 4.1 Soil-Moisture and Environmental Prediction Studies

Early investigations into soil-moisture estimation employed statistical and neural approaches that were primarily descriptive rather than operational. [Arif et al. \(2013\)](#) introduced two Artificial Neural Network (ANN) models for estimating evapotranspiration and soil moisture in paddy fields using limited meteorological data. Although their model achieved commendable predictive accuracy ( $R^2 = 0.80$  in training), it lacked integration with real-time control mechanisms. Similarly, satellite-based approaches such as [Ghazali et al. \(2019\)](#) and [Sholihah et al. \(2022\)](#) derived soil-moisture and salinity indices from Landsat imagery and NDVI-based vegetation mapping. These methods were suitable for large-scale environmental monitoring but constrained by coarse spatial resolution and delayed temporal updates, limiting their applicability for responsive irrigation management.

Advancements in hybrid and optimisation-based ML models significantly enhanced predictive precision. [Prasad et al. \(2018\)](#) utilised a hybrid ensemble of Extreme Learning Machine (ELM) with empirical mode decomposition to capture non-linear soil-moisture dynamics, while [Chatterjee et al. \(2020\)](#) applied an ANN optimised via a Flower Pollination Algorithm, achieving superior stability across climatic variations. However, both studies remained predictive in isolation, without translating results into actionable irrigation control. Meanwhile, [Babaeian et al. \(2021\)](#) fused multisensor remote-sensing data with automated ML (AutoML) to estimate root-zone soil moisture at fine spatial scales, and [Karthikeyan and Mishra \(2021\)](#) extended this concept by modelling multi-depth soil layers through XGBoost ensemble learning. Despite offering high spatial fidelity ( $R^2 > 0.80$ ), these works required satellite or UAV imagery, making them resource-intensive and unsuitable for real-time, field-level decision systems in developing agricultural settings.

Other contemporary developments explored deep learning for soil-sensing enhancement. [Kurtuluş et al. \(2022\)](#) reviewed the use of CNN-based models (AlexNet, ResNet, VGG16) for proximal soil sensing, achieving classification accuracies above 99%. Nonetheless, these frameworks primarily focused on sensor calibration and imaging rather than irrigation forecasting. Collectively, these studies reveal that while soil-moisture prediction has reached algorithmic maturity, real-time, closed-loop implementations remain scarce—an essential gap that the present

Smart Water Gate project aims to bridge through localised sensing, predictive modelling, and autonomous control.

## 4.2 IoT and Automation Applications

The proliferation of IoT technologies has led to a wide range of hardware-driven irrigation control systems. Early practical contributions include [Sanjula et al. \(2020\)](#), who proposed an automated water-gate control system for paddy fields using IoT-enabled sensors and PID-based water-level regulation. The system effectively maintained an 8 cm water level but lacked adaptive intelligence or dynamic scheduling. Building upon similar concepts, [Rama et al. \(2021\)](#) developed a fuzzy logic-based ESP32 water-gate controller capable of modulating gate openings in 50% increments based on soil and water-level readings. Although this introduced flexible rule-based control, it remained reactive and excluded predictive modelling. Likewise, [Desnanjaya and Nugraha \(2021\)](#) presented a web-based sluice gate system with Arduino and GSM integration, while [Niharika et al. \(2024\)](#) (GateGuard) incorporated GSM alerts and ThingSpeak cloud dashboards for remote farmer monitoring. Each of these implementations demonstrated feasibility in automating water control but failed to include machine-learning-based decision-making.

Within the Sri Lankan context, [Rathnayaka and Park \(2024\)](#) conducted one of the most comprehensive reviews of water-gate automation systems, outlining a closed-loop PID-based control architecture with solar-powered sensors and web monitoring. While this represented a significant step toward scalable automation, its reliance on rule-based logic limited adaptability to changing weather and crop phases. Furthermore, systems such as [Tolentino et al. \(2021\)](#) employed IoT with the Alternate Wetting and Drying (AWD) method, saving approximately 20% of irrigation water in Philippine paddy fields, yet were driven by fixed thresholds rather than intelligent prediction. Similarly, [Bwire et al. \(2024\)](#) empirically demonstrated the water-saving benefits of AWD, achieving 29% reduction compared to continuous flooding, but the implementation remained manual.

These automation-oriented studies collectively highlight a transition from manual irrigation to remote and rule-based control. However, they also reveal a persistent absence of predictive reasoning, adaptive logic, and integration of weather forecasting. The proposed Smart Water Gate system advances this line of research by merging low-cost IoT sensing, rain forecasting, and ML-based actuation to deliver a truly autonomous, data-driven control framework for paddy cultivation.

## 4.3 Crop Water Requirement and Forecasting Models

Crop Water Requirement (CWR) estimation has traditionally relied on static climatic data and empirical formulas. The FAO CROPWAT-8.0 framework [Singh and Verma \(2020\)](#) remains the

most widely used tool, employing the Penman–Monteith equation and fixed crop coefficients to compute monthly irrigation demand. Although reliable for macro-scale water planning, such models lack temporal resolution and responsiveness to microclimatic variability. [Awad et al. \(2021\)](#) compared empirical CROPWAT estimates with hydrological simulations using DRAINMOD, revealing discrepancies up to 59% and underscoring the uncertainty inherent in simplified empirical assumptions. While their proposed hydrology-driven approach improved theoretical accuracy, it remained simulation-based and devoid of real-time applicability.

Recent research has leveraged machine learning to overcome these limitations. [Nagappan et al. \(2025\)](#) evaluated LSTM, ANN, RF, and SVM models for CWR prediction, reporting an  $R^2$  of 0.92 for LSTM and demonstrating the power of temporal sequence modelling. However, the system was deployed via a web interface without hardware integration. Similarly, [Saikai et al. \(2023\)](#) introduced a Deep Reinforcement Learning (DRL) framework for irrigation scheduling using simulated wheat-field data from APSIM. The model autonomously learned profit-maximising irrigation strategies but was confined to virtual simulation, without real-world sensor feedback or control hardware. Complementary hydrological studies, such as [Arif et al. \(2013\)](#) and [Babaeian et al. \(2021\)](#), have further emphasised the relationship between soil moisture and evapotranspiration but again fell short of field-scale automation.

The reviewed literature thus reveals a methodological evolution—from empirical estimation to machine-learning-driven forecasting—but persistent gaps in operational deployment. Existing models largely serve as decision-support tools for human interpretation rather than self-regulating systems. The proposed project addresses this gap by uniting predictive CWR estimation with automated actuation through motorised water gates, bridging analytical prediction and practical implementation.

#### 4.4 Hybrid and AI-Driven Systems

Recent innovations have sought to integrate prediction, optimisation, and control into cohesive frameworks. [Togneri et al. \(2022\)](#) combined IoT-based sensor data with ensemble ML classifiers (RF, SVM, NB, XGBoost) to forecast soil moisture for irrigation. Despite its robustness, the system remained limited to valve control via relays and did not extend to hydraulic gate mechanisms required for paddy irrigation. [Nguyen et al. \(2022\)](#) advanced this direction by deploying ML models on edge devices such as Raspberry Pi to enable on-field inference and reduce latency. This approach aligns closely with the current project's emphasis on local computation, though it still lacked closed-loop actuation. Similarly, [Chatterjee et al. \(2020\)](#) and [Prasad et al. \(2018\)](#) introduced hybrid optimised algorithms that enhanced prediction stability but offered no hardware-level integration.

More advanced AI-driven designs have recently emerged. [Anbarasu and Parthipan \(2025\)](#) proposed an IoT-enabled Bee Colony Optimisation–ResNet hybrid model (BCO–ResNet) for

paddy irrigation, achieving 91.3% prediction accuracy and 38% water savings. However, its reliance on GPU-based cloud computation limited field deployability. Other studies, such as [Tolentino et al. \(2021\)](#) and [Rathnayaka and Park \(2024\)](#), adopted hybrid cyber-physical architectures integrating sensors, communication networks, and control units, yet still lacked adaptive or forecast-driven intelligence. Reinforcement-learning frameworks [Saikai et al. \(2023\)](#) further underscored the feasibility of intelligent irrigation scheduling but remained simulation-bound.

Collectively, the evolution of research from static modelling to intelligent control demonstrates clear progress but leaves notable gaps. The majority of systems either focus exclusively on prediction without automation or on automation without predictive intelligence. The proposed Smart Water Gate System distinguishes itself by integrating AI-based soil-moisture forecasting, rainfall prediction, and edge-controlled gate actuation into a unified, autonomous architecture. Unlike cloud-dependent or simulation-based systems, it performs computation locally on a Raspberry Pi, enabling resilience, low latency, and adaptability to Sri Lankan field conditions. This convergence of IoT sensing, AI reasoning, and mechatronic actuation establishes the groundwork for scalable, cost-effective smart irrigation tailored to paddy cultivation, addressing the long-standing divide between digital prediction and physical water management.

## 5 Identified Research Gaps and Justification

A synthesis of the reviewed literature reveals several consistent limitations that justify the proposed research direction.

1. **Lack of Real-Time Closed-Loop Integration** — Most studies are either predictive ([Arif et al. \(2013\)](#); [Nagappan et al. \(2025\)](#); [Babaeian et al. \(2021\)](#); [Karthikeyan and Mishra \(2021\)](#)) or automated ([Sanjula et al. \(2020\)](#); [Rathnayaka and Park \(2024\)](#); [Niharika et al. \(2024\)](#); [Rama et al. \(2021\)](#)) but rarely combine both aspects into a single real-time feedback system linking machine learning with actuation.
2. **Limited Forecast-Driven Intelligence** — Very few systems integrate rainfall or weather forecasts into irrigation scheduling ([Sanjula et al. \(2020\)](#)), relying instead on instantaneous sensor readings or historical averages. This limits the ability to anticipate changing water demands under variable climate conditions.
3. **Absence of Crop-Specific Adaptation** — Many existing frameworks ([Togneri et al. \(2022\)](#); [Nguyen et al. \(2022\)](#); [Saikai et al. \(2023\)](#)) are designed for general-purpose irrigation or non-paddy crops and fail to account for paddy's alternating wet–dry cultivation cycle, which requires dynamic water-level control.
4. **Non-Adaptive Control Logic** — Fuzzy and PID controllers respond reactively to sensor thresholds without learning from historical or forecasted data ([Rama et al. \(2021\)](#); [Desnanjaya and Nugraha \(2021\)](#); [Tolentino et al. \(2021\)](#)), preventing continuous optimi-

sation of irrigation decisions.

5. **Data-Scale Limitations** — Remote-sensing approaches ([Ghazali et al. \(2019\)](#); [Sholihah et al. \(2022\)](#)) offer wide spatial coverage but limited temporal resolution, while IoT systems ([Togneri et al. \(2022\)](#); [Nguyen et al. \(2022\)](#)) provide high-frequency data yet struggle with generalisation and long-term scalability.
6. **Manual Dependence and Lack of Automation** — Several existing systems, including [Niharika et al. \(2024\)](#); [Sanjula et al. \(2020\)](#); [Rama et al. \(2021\)](#), continue to rely on manual dashboard supervision, GSM-based notifications, or predefined control parameters requiring human oversight. For instance, [Niharika et al. \(2024\)](#) employed SMS alerts for farmers to trigger irrigation manually, while [Sanjula et al. \(2020\)](#) required user configuration despite using a PID controller. Similarly, the fuzzy logic model in [Rama et al. \(2021\)](#) depended on static rule sets that lacked adaptive learning. Such dependence on human intervention limits full autonomy and scalability in real-world agricultural environments.
7. **Limited Environmental Awareness** — No jointly consider the triad of soil moisture, tank water level, and rainfall forecast in a unified predictive model. This lack of environmental context limits comprehensive water management.

Addressing these gaps requires an integrated system that links environmental sensing, predictive analytics, and mechanical actuation through a unified CPS architecture. The proposed *Smart Water Gate System* contributes to the field by introducing an AI-driven, forecast-aware, and sensor-integrated irrigation framework specifically tailored for paddy cultivation. It advances existing work by transforming irrigation management from a reactive to a proactive, intelligent, and self-regulating process.

## 6 Chapter Summary

This chapter has critically reviewed the theoretical foundations, technologies, and empirical studies relevant to intelligent irrigation and paddy-field water management. It demonstrated that while substantial progress has been made in soil-moisture prediction, IoT-based sensing, and automation, integration across these domains remains limited. Existing systems are predominantly static, rule-based, or simulation-oriented, lacking adaptive intelligence and real-time feedback control.

The proposed research addresses these shortcomings through a novel closed-loop CPS framework that merges machine learning-based prediction, rainfall forecasting, and automated gate actuation. The insights from this review directly inform the research methodology presented in the following chapter, which details the system design, data-collection strategy, and model-development process.

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